

# Unit

1

## Lesson

2

# Spark The Challenge!

## EDP in Action!

Time Frame: [90 min]

### Lesson Objectives:

**By the end of this lesson students will be able to:**

- Define a real-world problem based on elderly mobility challenges
- Use empathy and research to better understand a users needs
- Identify design constraints related to the Smart Walker project
- Differentiate between roles of mechanical and robotics engineers
- Begin using the EDP as a guide for designing their own Smart Walkers

## Challenge questions of the lesson

- What is the Engineering Design Process, and why is it useful when solving real-life problems?
- How can we use empathy to define a meaningful and real-world problem?
- What are the specific challenges elderly people face when trying to walk safely or independently?
- What limitations or constraints do we need to consider when designing a smart walker?
- How can defining the problem well help us build a better solution?

## SEL Relation

- **Self-awareness & Empathy:**
  - Students learn to put themselves in another person's shoes, specifically an elderly person facing physical and emotional challenges.
  - They reflect on how it feels to depend on others or face barriers in daily tasks.
- **Relationship Skills & Teamwork:**
  - Students engage in respectful communication, active listening, and collaborative decision-making while defining the problem as a team.
- **Responsible Decision-Making:**
  - Students consider the real-world impact of their designs, showing care and responsibility toward a

## Engage



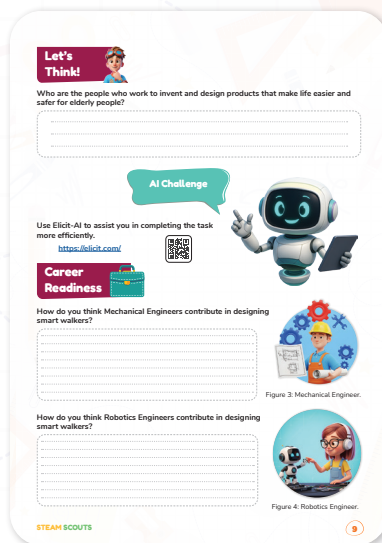
### Teacher's Role:

- **Reflective Coach:**
  - Prompt students to reflect on personal interest and strengths with the question: "Which job would YOU want to try—and why?"
  - Support students in articulating their thinking through sentence starters or journaling.
  - Provide opportunities for students to share their choices in pairs or small groups.
- **Facilitator:**
  - Lead an open discussion around how these professionals contribute to improving quality of life for the elderly.
  - Use guiding questions to help students make connections between the skills of each role and the features of assistive technologies.
- **Questioner:**
  - Pose a transition inquiry to spark curiosity: "Who are the people who work to invent and design products that make life easier and safer for elderly people?"
  - Encourage students to explore the human-centered purpose behind engineering careers.

### Expected From Students:

- Actively contribute to discussions by sharing thoughts, asking questions, and making thoughtful predictions based on the story.
- Express personal ideas confidently while listening respectfully to peers, fostering a collaborative and inclusive learning environment.
- Watch the videos actively, noting how each engineer contributes.
- Compare the roles:
  - Mechanical → Shape, structure, strength
  - Robotics → Sensors, programming, smart features
- Reflect and write (or discuss): "What feature of a Smart Walker needs both kinds of engineers?"

## Parts of the book included.



### Let's Think!



#### Objective:

Students will think critically about who designs products that help make life easier and safer for elderly people. This encourages students to recognize the roles of engineers, designers, and caregivers, and understand how these professions work together to solve real-world problems.

### Possible Strategies for implementation:

#### 1. Group Brainstorming:

- Have students discuss and brainstorm in small groups who they think plays a role in designing products for elderly people. Encourage them to list different job titles and roles, such as product designers, engineers, ergonomists, and caregivers.

#### 2. Research Activity:

- Ask students to research a specific profession (like an engineer or designer) and how that role impacts the creation of assistive technologies for the elderly. They can present their findings to the class.

#### 3. Role-Playing Activity:

- Have students role-play as different professionals (e.g., an engineer, a caregiver, a designer) and act out how each role contributes to solving challenges faced by elderly people.

#### 4. Class Discussion:

- After completing the activity, facilitate a class discussion on how these professionals work together. Ask questions like: "Why is teamwork important in designing solutions for elderly people?" or "How can engineers ensure that their designs meet the needs of elderly users?"

## 5. Empathy Mapping:

- Use an empathy map to explore the daily challenges of elderly people. Have students use this tool to design a product that helps address one of these challenges, taking into account the roles of different professionals.

### Career Readiness



#### Objective:

Students will explore the roles of Mechanical Engineers and Robotics Engineers in designing smart walkers. They will understand how each type of engineer contributes to the creation of assistive technology for elderly individuals.

#### Resources:

**Mechanical Engineer:** [https://www.youtube.com/watch?v=UrT1\\_TuvZmQ](https://www.youtube.com/watch?v=UrT1_TuvZmQ)  
<https://www.youtube.com/watch?v=JhzjIPvWG7Y>

**Robotics Engineer:** <https://www.youtube.com/watch?v=0dwkGhRPQW4>



#### Possible Answers:

- **Mechanical Engineers:**  
"Mechanical engineers help design the structure of the smart walker. They work on the frame, wheels, and handles, ensuring the walker is durable, lightweight, and comfortable for the user."
- **Robotics Engineers:**  
"Robotics engineers focus on integrating smart technology into the walker. They design the sensors, motors, and Bluetooth communication system to help the walker detect obstacles and assist with mobility."

### Possible Strategies for implementation:

#### 1. Watch & Discuss:

- Have students watch the provided videos about Mechanical Engineers and Robotics Engineers.
- After watching, ask students to discuss in pairs or small groups how each type of engineer contributes to the walker design.

#### 2. Group Brainstorming:

- In groups, ask students to brainstorm additional features that could be added to the Smart Walker based on the roles of each engineer.

#### 3. Role Play:

- Assign roles (Mechanical Engineer, Robotics Engineer, Designer) to students. Have them role-play a meeting where they discuss how they would work together to design a Smart Walker.

#### 4. Research Activity:

- Ask students to research real-world examples of products designed by Mechanical Engineers and Robotics Engineers. They can present how those designs helped people (e.g., robots for healthcare or mobility aids).

### Explain



#### Teacher's Role:

- **Guide students through the video:** Ensure that the students understand the Engineering Design Process (EDP) as it applies to their Smart Walker project.
- **Facilitate discussion:** After watching the video, help students reflect on how they can use the EDP steps (Define, Research, Ideate, Prototype, Test, and Improve) in their design work.
- **Encourage collaboration:** Remind students to share ideas within their teams and step into their roles as engineers and designers. Ensure that all students are actively participating in brainstorming and planning.

#### Expected From Students:

- **Watch the video:** Pay attention to the explanation of the Engineering Design Process (EDP).
- **Discuss in teams:** Collaboratively define the problem they're solving (e.g., improving mobility, safety, or independence for elderly people).
- **Use the EDP:** Apply the EDP steps to begin brainstorming and planning their Smart Walker design.
- **Document ideas:** Record their design process in notebooks or on digital platforms to refer back to as they move forward with their project.

#### Parts of the book included.



##### Screen Lessons

Now it's time to step into the role of engineers and use the Engineering Design Process (EDP).

[https://www.youtube.com/watch?v=MAhpfFt\\_mWM](https://www.youtube.com/watch?v=MAhpfFt_mWM)



##### Screen Lessons

#### Objective:

Students will understand and apply the Engineering Design Process (EDP) to their Smart Walker project. They will step into the role of engineers, defining problems, brainstorming solutions, and using the EDP to guide their design process.

Video Resources: [https://www.youtube.com/watch?v=MAhpfFt\\_mWM](https://www.youtube.com/watch?v=MAhpfFt_mWM)





## Possible Strategies for implementation:

### 1. Video Warm-Up + KWL Chart (Before, During, After Viewing)

**Strategy:** Activate prior knowledge and connect students to the topic.

**Implementation:**

- Before the videos, give students a KWL chart (Know, Want to Know, Learned).
- Ask:
  - What do you already know about how engineers solve problems?
  - What would you like to know about the steps engineers follow?
- After viewing, complete the "Learned" section together.

**Tip:** You can use a shared Google Doc or Jamboard for digital classrooms.

### 2. EDP Step Sorting Activity

**Strategy:** Reinforce understanding of the EDP steps.

**Implementation:**

- Provide students with step cards and example cards (based on the video examples).
- Example:
  - Card A: "Brainstorming different ways to carry a pizza"
  - Card B: "Drawing your best idea before building"
- Students match them to the correct step: Ask, Imagine, Plan, Create, Improve.

### 3. Think-Pair-Share: Why EDP Matters

**Strategy:** Help students reflect on why the EDP is useful in solving real-life problems.

**Implementation:**

- Ask students to reflect on the video prompt:
  - "Why do engineers need to follow a process like this?"
  - "What would happen if they skipped a step?"
- Pair up students to share thoughts.
- Then open a brief class discussion and chart out their responses.

### 4. Real-World Connection Writing Prompt

**Strategy:** Link the EDP to everyday life and your design challenge.

**Implementation:**

- Prompt students:
  - "Think of a time when something didn't work right — like a broken toy, bike, or app. How would you fix it using the Engineering Design Process?"

# Engineering Design Process

Name: .....

Project: .....

## Ask

- Identify the Problem  
What specific challenge or need are you trying to solve?  
For example: How can we make walking safer and easier for elderly people?
- Understand the Requirements  
What are the essential features your solution needs?  
For example: The walker must be lightweight, stable, and easy to use.



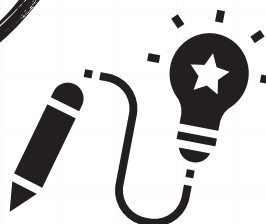
## Imagine

- Brainstorm Solutions  
Think of as many ideas as possible. Don't limit yourself—be creative!  
For example: Could we add sensors to the walker to detect obstacles or use a vibration motor to alert the user?
- Consider Multiple Approaches  
What different materials or technologies could be used to solve this problem?  
For example: Should we use an ultrasonic sensor or an infrared sensor for obstacle detection?



## Plan

- Select the Best Solution  
Choose the idea that best meets the requirements and constraints.  
For example: We'll use an ultrasonic sensor for obstacle detection because it works well in different lighting conditions.
- Create a Design  
Draw sketches or diagrams of your idea, and think about how all the components will work together.  
For example: Sketch the Smart Walker frame, and where to place the sensor, motor, and control button.
- Prepare Materials & Resources  
Plan out what materials you'll need and how you will test your design.  
For example: We need PVC pipes for the frame, an ultrasonic sensor, a vibration motor, and an Arduino board.



## Create



## Improve







### Objective:

Students will reflect on their understanding of mobility challenges faced by elderly individuals and how empathy, engineering roles, and the Engineering Design Process (EDP) contribute to designing assistive solutions like the Smart Walker. They will identify key design constraints, differentiate between mechanical and robotics engineering roles, and articulate the first steps in planning their own user-centered innovation.

### Activity Instructions:

#### Step 1: Revisit the lesson

Students review the lesson materials and activities, including empathy maps, videos, observations, and any initial sketches or notes related to the Smart Walker.

#### They should reflect on:

- What specific mobility problems do elderly individuals face in real life?
- What did I learn through empathy, stories, or research that helped me understand their needs?
- What design constraints (e.g., safety, weight, ease of use) must I consider for my Smart Walker?
- What roles do mechanical and robotics engineers play in solving these kinds of problems?
- Where am I in the Engineering Design Process, and how have I started applying it?

#### Step 2: Student Self-Reflection

Students individually reflect on their understanding and design thinking process. Prompts may include:

- “One real-world problem I now understand better is...”
- “Empathy helped me realize that elderly people need...”
- “An important constraint I need to keep in mind is...”
- “Mechanical engineers focus more on ...., while robotics engineers focus more on .....”
- “Here’s how I started planning my Smart Walker using the Engineering Design Process...”

They may record answers in journals, on digital slides, or using sentence frames.

#### Step 3: Small Group Discussion

In small groups, students share one insight from their self-reflection.

Use Think-Pair-Share or Round Robin to ensure equal participation.

#### Discussion questions:

- “What’s one need your Smart Walker design must address?”
- “What’s a challenge you faced while applying the EDP?”
- “What part of designing with empathy was hardest—and why?”

- “Which engineering role are you most interested in, and how would you contribute to this project?”

## Relation to Standards

### 1. NGSS (Next Generation Science Standards)

- **MS-ETS1-1** – Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution.
- **MS-ETS1-2** – Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
- **MS-ETS1-3** – Analyze data from tests to determine similarities and differences among several design solutions.
- **MS-ETS1-4** – Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process.

### 2. ISTE Standards for Students

- **1.1 Empowered Learner:** Students articulate and set personal learning goals, develop strategies to leverage technology, and reflect on the learning process.
- **1.4 Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions.
  - **1.4.a:** Know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems.
  - **1.4.b:** Select and use digital tools to plan and manage a design process that considers design constraints and calculated risks.
  - **1.4.c:** Develop, test and refine prototypes as part of a cyclical design process.
- **1.7 Global Collaborator:** Students use digital tools to broaden their perspectives and enrich their learning by collaborating with others and working effectively in teams.

### 3. Common Core State Standards (CCSS) – ELA & Literacy in Science and Technical Subjects

- **CCSS.ELA-LITERACY.RST.6-8.1:** Cite specific textual evidence to support analysis of science and technical texts.
- **CCSS.ELA-LITERACY.RST.6-8.3:** Follow precisely a multistep procedure when carrying out experiments, taking measurements, or performing technical tasks.
- **CCSS.ELA-LITERACY.WHST.6-8.7:** Conduct short research projects to answer a question (including a self-generated question), drawing on several sources and generating additional related, focused questions for further research and investigation.

- **CCSS.ELA-LITERACY.SL.7.1:** Engage effectively in a range of collaborative discussions with diverse partners on grade 7 topics, texts, and issues, building on others' ideas and expressing their own clearly.